

Muskan Yadav

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Summary

Data Analyst — AI Enthusiast — Integrated M.Tech (CSE – Computational & Data Science), VIT Bhopal (CGPA: 7.98/10). Skilled in Python, SQL, Power BI, Machine Learning, Data Analytics, Data Visualization, Excel, and AI. Built 3+ end-to-end AI & software projects, transforming data into actionable insights and developing scalable, data-driven solutions.

Education

Vellore Institute of Technology, Bhopal

2022 – 2027

Integrated M.Tech (B.Tech + M.Tech) in CSE — CGPA: 7.98/10

Experience

Data Science Intern

Dec 2024 – Feb 2025

CodeSoft

- Developed machine learning pipelines using Scikit-learn; solved 5+ real-world business problems through iterative model tuning.
- Performed data analysis and manipulation using Pandas and NumPy across multiple structured datasets to surface key trends.
- Executed ETL workflows — preprocessing and feature engineering — across 4+ datasets; improved data accuracy by 15%.
- Validated data quality and consistency, ensuring reliable inputs for downstream analysis and modeling.
- Created statistical predictive models that informed data-driven business decision-making.
- Crafted Matplotlib and Excel visualizations to communicate insights; cut manual reporting time by 20%.

Technical Skills

Programming Languages: Python, SQL

Data Science & ML: Power BI, Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, SARIMA, Prophet, PyTorch, LSTM, Time Series Forecasting, EDA, Predictive Analytics

AI & Generative AI: LLMs, RAG, AI Agents, Groq API, Llama 3

Databases: MySQL, PostgreSQL, MongoDB

Developer Tools: Git, GitHub, Docker, MLflow, VS Code, Jupyter Notebook, Postman

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks

Projects

Retail Demand Forecasting & Inventory Optimization | *Python, XGBoost, Prophet, PyTorch, FastAPI, Streamlit, Docker* | GitHub

- Engineered a forecasting engine on 3.5 years of retail data across 25 store-department segments; benchmarked 4 architectures (SARIMA, Prophet, LSTM, XGBoost).
- Achieved 4.74% MAPE using global XGBoost with lagged/rolling features, outperforming Prophet (4.93%) and LSTM (10.37%).
- Optimized EOQ, Safety Stock, and ROP parameters; achieved 15–20% cost savings in a 26-week simulation.
- Deployed 2 microservices (FastAPI + Streamlit) via Docker Compose; tracked experiments with MLflow.

CarpoolPal – Ride Sharing Application | GitHub

- Built a full-stack carpooling platform using FastAPI, MongoDB, and React.js with 8+ REST endpoints.
- Trained a Random Forest model (100 estimators) on 1,000+ ride records, achieving fare predictions accurate to 12/km.
- Applied the Haversine Formula and secured auth with JWT/bcrypt; improved peak-hour fare accuracy by 18%.

MedAgent — AI Medical Assistant | *Python, Groq API, Llama 3, RAG, Streamlit* | GitHub

- Architected an agentic AI medical assistant using RAG and the Llama 3 LLM to answer health queries from medical documents.
- Optimized inference using Groq's LPU-based architecture; reduced response time to under 2 seconds per query, a 4x speedup.
- Enabled real-time RAG-based retrieval across 100+ medical topics, improving query coverage and accuracy.
- Maintained multi-turn conversational context using session memory across 10+ exchanges.

Certificates

Artificial Intelligence Fundamentals – IBM SkillsBuild (2024)

Applied Machine Learning in Python – Coursera (2023)

HackerRank 5-Star in Python – Solved 100+ algorithmic problems

Leadership / Achievements

Volunteered with the NSS Club, leading 5+ awareness drives and coordinating 3+ campus events.

Kanchan Education and Welfare Society: Supported welfare programs reaching 50+ beneficiaries.